THE EQUALIZER OF TWO MORPHISMS IS A LOCALLY CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4326-4339, following the statement at 4316-4324.

Editorial note. The categorical identification 'the equalizer is the fibre product for categorical reasons' is anchored by the displayed diagram.

Allowed standard prerequisites (editorial).

- Immersions stable under base change. lemma-base-change-immersion (card L26430 this batch, cited at ~4337). Z -> X is an immersion as a base change of the diagonal.
- The relative diagonal is an immersion. lemma-diagonal-immersion (card L26408, cited at ~4337). Supplies that Delta_{Y/S} is an immersion.
- Definition of separated morphisms. definition-separated (same chapter, cited at ~4338). Closedness of Z when Y/S is separated.

Source reference. Original definition 01KK: sources/stacks-schemes/pinned-original-source.tex, source lines 4279–4279. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JY: sources/stacks-schemes/pinned-original-source.tex, source lines 3662–3662. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KJ: sources/stacks-schemes/pinned-original-source.tex, source lines 4255–4255. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X, Y be schemes over S. Let $a, b : X \to Y$ be morphisms of schemes over S. There exists a largest locally closed subscheme $Z \subset X$ such that $a|_Z = b|_Z$. In fact Z is the equalizer of (a, b). Moreover, if Y is separated over S, then Z is a closed subscheme.*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The equalizer of (a, b) is for categorical reasons the fibre product Z in the following diagram

$$\begin{array}{ccc} Z = Y \times_{(Y \times_S Y)} X & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle (a,b)} \\ Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y \end{array}$$

Thus the lemma follows from Lemmas 01JY, 01KJ and Definition 01KK. $\square$